

π -Diff: Physically-Inspired Low-Light Image Enhancement with Structure Preserving Diffusion Priors

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Abstract

Low-Light Image Enhancement (LLIE) aims to improve the visual quality of images captured under insufficient illumination. While recent diffusion model (DM)-based approaches have shown promising performance, they typically couple denoising with illumination and color restoration. This leads to high prediction uncertainty, generating images with blurred details, under-exposure, color distortion, and hallucinated content. To address these issues, we propose π -Diff, an unsupervised, training-free framework that incorporates a **Physically-Inspired** enhancement process utilizing structure-preserving **Diffusion** priors. Breaking the conventional direct-generation paradigm, we reinterpret the reverse diffusion output as a noiseless structural proxy of the low-light input, preserving essential details for subsequent enhancement. Based on this refined proxy, we introduce a physically-explainable enhancement strategy. Specifically, during the diffusion sampling, we apply an ISP-inspired recovery process to the decomposed illumination map by dynamically estimating an adaptive light gain to compensate for low-photon condition and further introducing a gamma corrector for exposure remapping. Meanwhile, the decomposed reflectance map is constrained to preserve the consistency of scene's intrinsic color and content. Extensive experiments on publicly available real-world benchmarks demonstrate that our method achieves exceptional performance compared to existing approaches.

1. Introduction

Images captured in low-light environments frequently exhibit various degradations, including severe noise, color distortion, and reduced visibility, which significantly impair their utility in downstream applications, such as autonomous driving and nighttime surveillance. Consequently, Low-Light Image Enhancement (LLIE) has be-

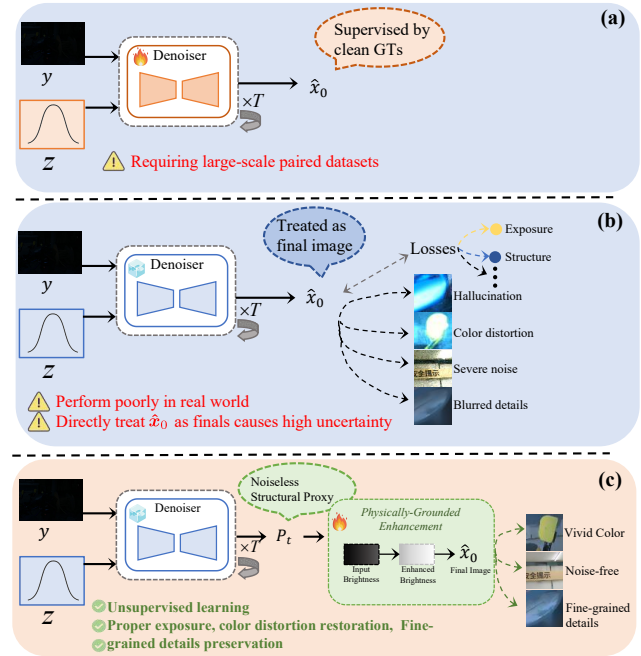


Figure 1. **Comparisons of existing DM-based methods.** (a) Supervised methods require extensive paired data for training; (b) Unsupervised methods follow a “direct-generation” paradigm by using reverse diffusion outputs as final results, bringing high prediction uncertainty; (c) Our method treats diffusion outputs as noiseless structural proxies and applies physics-inspired enhancement for superior results.

come a fundamental research area, aiming to restore degraded images to their normal-light counterparts.

Traditional LLIE techniques primarily relied on hand-crafted priors, such as gray transform and histogram equalization. However, their dependence on manually designed priors tailored for specific degradation patterns resulted in limited generalization and insufficient robustness across diverse low-light conditions. The emergence of deep learning has led to numerous learning-based approaches [12, 21, 25, 27, 29, 33, 55] that leverage advanced network archi-

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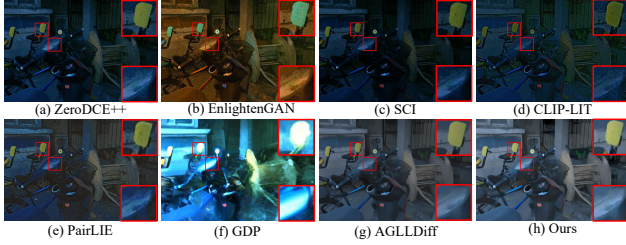


Figure 2. Visual comparisons of existing LLIE methods.

textures and training strategies to learn mappings from low-light to well-exposed images. Despite remarkable progress in LLIE, these learning-based methods often suffer from overfitting, producing undesirable visual artifacts such as under-enhancement, low contrast, and color distortion, as demonstrated in Fig. 2 (a)-(e).

Recently, diffusion models (DMs) [8, 19, 43, 47] have emerged as promising solutions for LLIE due to their exceptional capabilities in realistic image generation and detail preservation. However, existing DM-based methods relying on the supervised learning paradigm require large amounts of paired low/normal-light images for training (as illustrated in Fig. 1(a)). Collecting such paired data in practice is expensive, challenging, and prone to misalignment. To avoid the dependency on paired datasets, some unsupervised methods [10, 28] have introduced pre-trained diffusion models as priors for zero-shot image enhancement. Yet, they typically follow a “direct-generation” paradigm (Fig. 1(b)). For example, GDP [10] adopts a simple degradation model to simulate low-light inputs and directly assumes the DM’s denoised outputs as the final enhanced results. Likewise, AGLLDiff [28] imposes constraints on the DM’s denoised output to resemble normal-light images. This paradigm tightly couples denoising, illumination and color distortion restoration within the same diffusion sampling by directly treating the reverse diffusion outputs as the final clean images. Since these distinct physical processes are mixed together, the reverse sampling faces high prediction uncertainty. As shown in Fig. 1(b) and Fig. 2(f-g), these models often generate images with blurred details, residual noise, and color distortion, and even hallucinate non-existent textures, failing to generate high-quality images.

In this paper, we propose π -Diff, an unsupervised, training-free framework that incorporates a **Physics-Inspired** enhancement strategy utilizing pre-trained **Diffusion** priors while preserving structure details. Unlike previous works that directly treat the reverse diffusion output as the final result, we reinterpret it as a noiseless structural proxy of the low-light input. This crucial paradigm shift decouples the denoising process from illumination adjustment and color distortion restoration, fundamentally mitigating high prediction uncertainty and

preserving essential fine-grained details for subsequent processing. Building upon this structural representation, we design a jointly optimized physical ISP-inspired recovery process for the enhancement of the decomposed illumination map and constrain the corresponding reflectance map to preserve color consistency. The contributions are summarized as follows:

- We propose π -Diff, an unsupervised, training-free framework that effectively leverages diffusion priors to restore normal-light images from low-light inputs without paired training data.
- Breaking the conventional direct-generation paradigm, we introduce a novel noiseless structural proxy to mitigate the blurred artifacts, preserving fine-grained structural details.
- We introduce a physically-explainable enhancement strategy featuring an ISP-inspired illumination recovery. Utilizing pre-trained diffusion priors, we jointly estimate an adaptive light gain to compensate for low-photon condition and a non-linear gamma corrector for optimal exposure remapping during the diffusion sampling process.
- Comprehensive experiments demonstrate that our method achieves exceptional performance, generating images with superior quality.

2. Related Works

2.1. Low-Light Image Enhancement

Low-light image enhancement has been a long-standing research area, as transforming illumination-degraded images into visually appealing, normal-light counterparts is crucial for numerous downstream applications, including object detection [18], 3D imaging [51, 52], and autonomous driving [26]. Early works relied on handcrafted priors, such as histogram equalization (HE) [3] and Retinex theory [23]. HE-based methods [35] enhance contrast by redistributing pixel intensities, while Retinex-based approaches [11, 16] decompose the image into reflectance and illumination components, subsequently adjusting the dynamic range of the illumination map to improve visual quality.

The advent of deep learning has revolutionized LLIE, with learning-based methods demonstrating superior robustness and visual quality compared to traditional techniques. These modern approaches can be categorized into supervised [31, 59, 63, 65] and unsupervised [12, 15, 21, 25, 29, 33] paradigms. Supervised methods leverage advanced network architectures and sophisticated training strategies to achieve superior enhancement quality. However, their reliance on large-scale paired datasets limits their practical applicability and generalization to diverse real-world degradations. Unsupervised methods, on the other hand, circumvent the need for paired data through techniques such as adversarial training [21] or iterative curve estima-

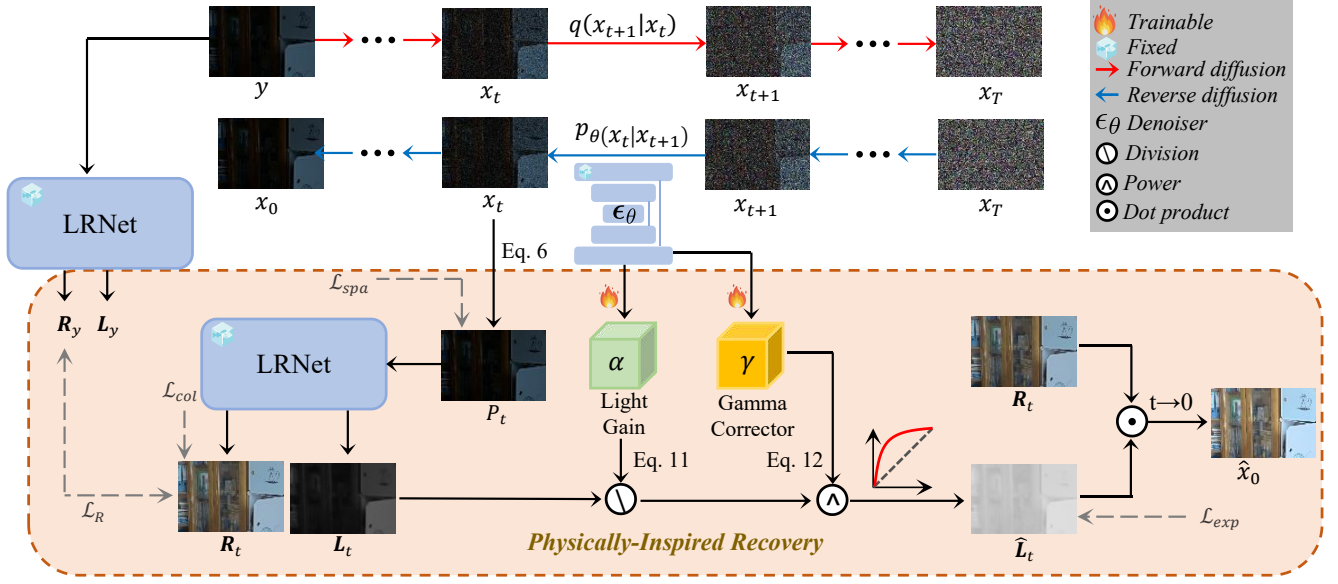


Figure 3. **Overview of the proposed π -Diff framework.** At each timestep t of the reverse diffusion process, we first derive a noiseless structural proxy P_t from the current noisy state x_t . Subsequently, the P_t is decomposed into a reflectance map R_t and an illumination map L_t via an off-the-shelf decomposition network $\text{LRNet}(\cdot)$. The proposed ISP-inspired recovery strategy optimizes an adaptive light gain α and a gamma corrector γ to enhance L_t . As $t \rightarrow 0$, the securely preserved reflectance R_0 and the properly exposed illumination $\hat{L}_0$ are multiplied to output the final enhanced image $\hat{x}_0$.

tion [15], achieving improved generalization. Nevertheless, their limited generative capacity often results in suboptimal enhancements characterized by residual noise, color distortion, and improper exposure adjustment.

2.2. Diffusion-Based Image Restoration

While numerous image restoration methods [36–42] have been proposed, diffusion models have recently emerged as a powerful paradigm for super-resolution [13, 46, 56], underwater image enhancement [48, 64], dehazing [22, 61], inpainting [1, 30], and low-light image enhancement [10, 28, 54]. Their success stems from their exceptional ability to generate high-quality images through iterative denoising processes. However, supervised diffusion-based methods come with significant practical limitations: they typically require extensive paired data, which is expensive and challenging to collect in real-world scenarios.

To address these limitations, recent research has focused on performing image restoration in an unsupervised manner by leveraging priors from diffusion models pretrained on large-scale natural image datasets. For instance, GDP [10] employs a simple linear degradation model to simulate low-light conditions and directly treats the DM’s denoised outputs as the final enhanced results. Quadprior [54] extracts illumination-invariant information from input normal-light images and reconstructs enhanced images using Stable Diffusion’s generative capability. Similarly, AGLLDiff [28] constrains the DM’s output to resemble normal-light im-

ages through explicit guidance. However, these methods either rely on oversimplified degradation models that fail to capture real-world lighting complexity, or directly interpret the reverse diffusion output as the final enhanced image, leading to blurred details and hallucinated content.

3. Method

3.1. Preliminary: Diffusion Models

The Denoising Diffusion Probabilistic Model transforms data distribution $x_0 \sim p_{data}$ into Gaussian distribution $x_T \sim \mathcal{N}(0, \mathbf{I})$ through two key processes: the forward process and the reverse process.

The forward process progressively corrupts training data by adding Gaussian noise according to a predefined schedule $x_1, x_2, \dots, x_T$:

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t\mathbf{I}), \quad (1)$$

where $x_1, x_2, \dots, x_T$ are noise latent variables. $\mathcal{N}$ represents the Gaussian distribution. Applying the reparameterization trick allows this process to be rewritten in closed-form as:

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)\mathbf{I}), \quad (2)$$

where $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{i=0}^t \alpha_i$.

The reverse process converts an isotropic Gaussian distribution back into the target data distribution by progressively denoising x_T until a high-quality image is produced.

Starting from Gaussian noise $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, the reverse process is defined as:

$$p_\theta(\mathbf{x}_0, \dots, \mathbf{x}_{T-1} | \mathbf{x}_T) = \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t), \quad (3)$$

Each denoising step can be further computed by approximating the posterior distribution as:

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \boldsymbol{\Sigma}_\theta \mathbf{I}), \quad (4)$$

where variance $\boldsymbol{\Sigma}_\theta \mathbf{I}$ is either fixed or learnable. The mean $\boldsymbol{\mu}_\theta(\mathbf{x}_t, t)$ is estimated by a denoising network $\epsilon_\theta(\mathbf{x}_t, t)$ according to:

$$\boldsymbol{\mu}_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \alpha_t}} \epsilon_\theta(\mathbf{x}_t, t) \right). \quad (5)$$

Typically, $\tilde{\mathbf{x}}_0$ can be derived from $\mathbf{x}_t$ in practice as:

$$\tilde{\mathbf{x}}_0 = \frac{\mathbf{x}_t}{\sqrt{\alpha_t}} - \frac{\sqrt{1 - \alpha_t} \epsilon_\theta(\mathbf{x}_t, t)}{\sqrt{\alpha_t}}. \quad (6)$$

Classifier guidance. For conditional generation, the classifier guidance [8] steers the unconditional diffusion model towards condition $\mathbf{y}$ by modifying the reverse process with a gradient term $\mathbf{g}$, defined as:

$$\mathbf{g} := \nabla_{\mathbf{x}} \log p_\phi(\mathbf{y} | \mathbf{x})|_{\mathbf{x}=\boldsymbol{\mu}_\theta(\mathbf{x}_t, t)}, \quad (7)$$

which adjusts the sampling distribution as:

$$p_{\theta, \phi}(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{y}) \approx \mathcal{N} \left(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t) + \boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t) \mathbf{g}, \boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t) \right). \quad (8)$$

3.2. Overview of π -Diff

Fig. 3 illustrates the overall pipeline of our proposed π -Diff for training-free low-light image enhancement. Given a low-light image $\mathbf{y}$ characterized by low visibility, severe noise, and color distortion, our objective is to restore a high-quality, well-lit image by leveraging the generative priors of a pre-trained diffusion model. At each timestep of the reverse diffusion process, we first extract a noiseless structural proxy from the current noisy image. Building upon this proxy, we introduce a physically-explainable enhancement strategy, where the decomposed reflectance map from the proxy is preserved for color and content consistency, and the illumination map undergoes an ISP-inspired recovery process. Finally, we design several loss functions to guide the diffusion sampling trajectory towards a well-exposed and structurally faithful output.

3.3. Noiseless Structural Proxy Formulation

In low-light conditions, high-frequency structures such as edges and fine textures are corrupted by noise. Conventional unsupervised diffusion-based methods attempt to tackle denoising and illumination recovery simultaneously through a direct-generation paradigm. However, as intuitively visualized in the Sobel edge maps in Fig. 5, existing methods struggle to preserve high-frequency details, leading to distorted and over-smoothing structures.

To understand why these direct-generation methods inevitably compromise structural details, we analyze the inherent statistical objective of the reverse diffusion process. According to minimum mean square error (MMSE) estimation theory [49], it is well-established that the optimal MSE denoiser predicts the conditional posterior mean of the data. Furthermore, based on Tweedie’s formula [6, 9], the network’s raw clean prediction $\tilde{\mathbf{x}}_0$ at any arbitrary timestep t (as formulated in Eq. 6) inherently approximates this conditional expectation:

$$\tilde{\mathbf{x}}_0 \approx \mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t, \mathbf{y}] = \arg \min_z \mathbb{E}[\|\mathbf{x}_0 - \mathbf{z}\|^2 | \mathbf{x}_t, \mathbf{y}] \quad (9)$$

where $\mathbf{y}$ denotes the given low-light input, $\mathbf{x}_t$ represents the noisy image at timestep t , and $\mathbf{x}_0$ is the ideal normal-light image. $\mathbf{z}$ serves as a placeholder for any possible generated image and $\tilde{\mathbf{x}}_0$ is the final enhanced image. This equation indicates that the raw prediction $\tilde{\mathbf{x}}_0$ is the specific image $\mathbf{z}$ that minimized the MSE against the target $\mathbf{x}_0$.

However, LLIE is a highly ill-posed problem. Given a single severe low-light observation $\mathbf{y}$, the true distribution of corresponding normal-light images $p(\mathbf{x}_0 | \mathbf{y})$ is highly complex and multi-modal, containing multiple plausible solutions with varying brightness levels and color adjustments. When calculating the expectation $\mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t, \mathbf{y}]$, the network is forced to compute the weighted average across all these distinct modes. As widely observed in generative modeling [44, 45], averaging diverging pixel values inevitably results in the degradation of high-frequency information. Therefore, if a method directly treats this raw prediction as the final enhanced result (i.e., setting $\hat{\mathbf{x}}_0 = \tilde{\mathbf{x}}_0$), it theoretically explains the blurry edges and hallucinated artifacts observed in existing methods.

Based on the above analysis, we propose a paradigm shift by repurposing the powerful diffusion prior exclusively as a robust structural extractor rather than a direct image generator. Instead of treating the prediction $\tilde{\mathbf{x}}_0$ as the final enhanced image $\hat{\mathbf{x}}_0$, we formally redefine it as our noiseless structural proxy, denoted as $\mathbf{P}_t$ (i.e., $\mathbf{P}_t = \tilde{\mathbf{x}}_0$).

This proxy decouples structural denoising from the comprehensive restoration. While the expectation operation blurs the multi-modal variations, it averages out the zero-mean random noise while retaining the statistically structures. By relying on the generative prior solely for noise

removal, P_t successfully strips away the severe low-light noise. Consequently, it establishes a reliable base for our subsequent physically-grounded enhancement.

3.4. ISP-Inspired Recovery

Despite providing a noise-free structural proxy, P_t still suffers from under-exposure. To address this, we first decompose P_t into a reflectance map and an illumination map:

$$R_t, L_t = \text{LRNet}(P_t), \quad (10)$$

where R_t represents the intrinsic properties of the scene, while L_t captures the degraded illumination. $\text{LRNet}(\cdot)$ is an off-the-shelf decomposition network based on the Retinex theory [58]. To prevent color distortion, we constrain the reflectance map R_t to maintain color constancy via designed losses in Sec. 3.5.

With the intrinsic colors of the scene preserved, we perform a learnable physical enhancement exclusively on the illumination map L_t . While the low-light inputs reside in the sRGB domain, their severe under-exposure fundamentally originates from the physical constraints of the camera’s image formation process under extreme photon attenuation. Rather than relying on black-box neural networks to directly map degraded images to normal-light targets, we propose an ISP-inspired cascade mechanism applied directly to L_t , which mathematically simulates the essential ISO amplification and non-linear tone mapping inherent in a real-world camera pipeline.

First, to physically compensate for the low-photon deficit, we introduce an adaptive light gain parameter α . We divide the illumination by this learnable gain, functioning analogously to applying a digital gain:

$$\hat{L}_t \leftarrow \frac{L_t}{\max(\epsilon, \alpha)}, \quad (11)$$

where ϵ is a small constant introduced to prevent zero-division instability.

However, applying light gain alone inevitably causes over-exposure in locally bright regions. We further introduce a non-linear gamma corrector γ for optimal exposure remapping and contrast balancing:

$$\hat{L}_t \leftarrow \hat{L}_t^{\frac{1}{\gamma}}, \quad (12)$$

This non-linear operation dynamically suppresses over-exposed highlights while revealing fine-grained details in extremely dark regions. Through this ISP-inspired recovery, we obtain the physically restored illumination map $\hat{L}_t$. By decoupling the structural denoising from this explicit illumination calibration, our pipeline successfully derives a perfectly exposed and structurally sharp foundation without disrupting the intrinsic scene colors. The final restored image $\hat{x}_0$ is obtained as timestep t approaches zero:

$$\hat{x}_0 = R_0 \cdot \hat{L}_0, \quad (13)$$

Algorithm 1 The sampling process of π -Diff

Input: Low-light image y , gradient scale s , light gain α , gamma corrector γ , learning rate l , spatial consistency loss $\mathcal{L}_{spa}$, reflectance consistency loss $\mathcal{L}_R$, exposure control loss $\mathcal{L}_{exp}$, color loss $\mathcal{L}_{col}$, decomposition network $\text{LRNet}(\cdot)$, diffusion model $(\mu_\theta, \Sigma_\theta)$

Output: Enhanced image $\hat{x}_0$

Sample $x_T \sim \mathcal{N}(0, I)$ **for** $t = T$ **to** 1 **do**

```

     $\mu, \Sigma \leftarrow \mu_\theta(x_t), \Sigma_\theta(x_t)$ 
     $\tilde{x}_0 \leftarrow \frac{x_t}{\sqrt{\alpha_t}} - \frac{\sqrt{1-\alpha_t} \cdot \epsilon_\theta(x_t, t)}{\sqrt{\alpha_t}}$ 
     $P_t \leftarrow \tilde{x}_0$  Noiseless structural proxy
     $R_y, L_y \leftarrow \text{LRNet}(y)$ 
     $R_t, L_t \leftarrow \text{LRNet}(P_t)$ 
     $\hat{L}_t \leftarrow \frac{L_t}{\max(\epsilon, \alpha)}$ 
     $\hat{L}_t \leftarrow \hat{L}_t^{\frac{1}{\gamma}}$ 
     $\mathcal{L}_{P_t} \leftarrow \mathcal{L}_{spa}(y, P_t) + \mathcal{L}_R(R_y, R_t) + \mathcal{L}_{col}(y, P_t)$ 
     $\alpha \leftarrow \alpha - l \nabla_\alpha \mathcal{L}_{exp}$ 
     $\gamma \leftarrow \gamma - l \nabla_\gamma \mathcal{L}_{exp}$ 
    Sample  $x_{t-1} \sim \mathcal{N}(\mu + s \nabla_{P_t} \mathcal{L}_{P_t}, \Sigma)$ 

```

return $\hat{x}_0 \leftarrow R_0 \cdot \hat{L}_0$

3.5. Physically-Guided Joint Optimization

To achieve zero-shot enhancement without requiring any paired training data, we formulate a set of loss functions to guide the reverse diffusion process. As detailed in Algorithm 1, our framework performs a two-fold joint optimization at each timestep t : guiding the structural proxy P_t for structural fidelity, and optimizing the learnable physical parameters $\{\alpha, \gamma\}$ for illumination adjustment.

Structural Guidance for the Proxy. To ensure that the diffusion sampling trajectory adheres to the original scene contents and intrinsic colors, we apply three constraints directly to the noiseless structural proxy P_t . First, we employ a spatial consistency loss between P_t and input y :

$$\mathcal{L}_{spa} = \|P_t - y\|_2^2, \quad (14)$$

Exposure Guidance for Illumination Recovery Parameters. We design an exposure control loss to dynamically search for the optimal light gain α and gamma corrector γ . Specifically, the restored illumination map $\hat{L}_t$ is average-pooled over non-overlapping patches, and the loss forces each patch’s illuminance toward an ideal well-exposed target e :

$$\mathcal{L}_{exp} = \frac{1}{N} \sum_{i=1}^N (m_i - e)^2, \quad (15)$$

where m_i is the average illuminance of the i -th patch, N denotes the total number of patches, and e is the ideal bright-

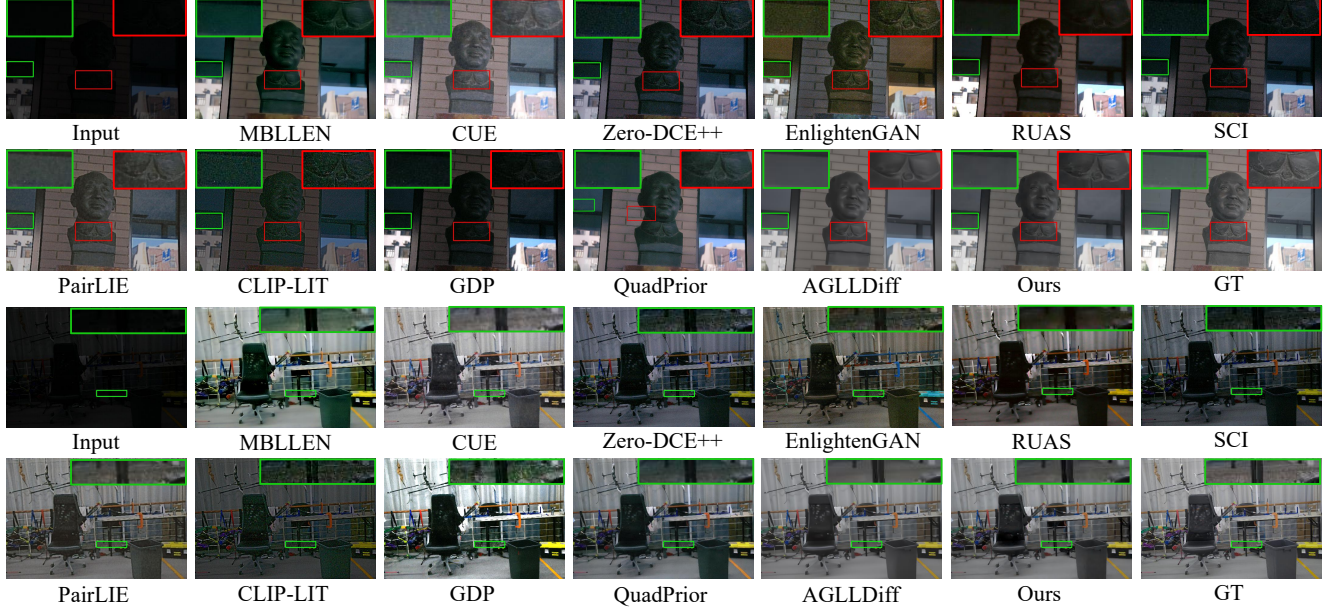


Figure 4. **Qualitative comparison on LOLv2-real captured dataset.** Our method effectively recovers detailed structure and natural appearance while other methods often struggle with detail preservation, denoising, color restoration, and illumination improvement.

ness scalar (set to 0.7 empirically). At each timestep, gradients derived from $\mathcal{L}_{exp}$ are back-propagated to iteratively update α and γ , explicitly driving the ISP-inspired recovery mechanism.

Color Guidance for Reflectance Map Restoration. To prevent severe color distortions, we introduce a reflectance consistency loss. Physically, the intrinsic reflectance of objects remains invariant under varying environmental illumination. Therefore, we enforce consistency between the reflectance map decomposed from the current proxy ($\mathbf{R}_t$) and that decomposed directly from the input image ($\mathbf{R}_y$):

$$\mathcal{L}_R = \|\mathbf{R}_t - \mathbf{R}_y\|_2^2, \quad (16)$$

where $\mathbf{R}_y$ represents the reflectance map decomposed from the low-light input $\mathbf{y}$, i.e., $\mathbf{R}_y, \mathbf{L}_y = \text{LRNet}(\mathbf{y})$.

Furthermore, we impose a color constancy constraint based on the Gray-World assumption:

$$\mathcal{L}_{col} = \sqrt{\sum_{p,q} (\mathbf{R}_t^p - \mathbf{R}_t^q)^2}, \quad (17)$$

where $(p, q) \in \{(R, G), (G, B), (B, R)\}$ represents any pair of color channels.

The total loss function guiding the overall network is the sum of these constraints:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{spa} + \lambda_2 \mathcal{L}_{exp} + \lambda_3 \mathcal{L}_R + \lambda_4 \mathcal{L}_{col}. \quad (18)$$

where $\lambda_1, \lambda_2, \lambda_3$ and λ_4 are the weighting coefficients for each loss function term.

4. Experiment

4.1. Experimental Settings

Datasets. To verify the effectiveness of our method, we evaluate on the public LLIE datasets, including LOLv1 [59], LOLv2-synthetic [59], LOLv2-real captured [60], LSRW-Huawei [17], and LSRW-Nikon [17]. Moreover, we further evaluate the generalization ability of the proposed method on five commonly used real-world unpaired benchmarks, including DICM [24], LIME [16], MEF [32], NPE [53], and VV [50].

Baselines. We compare our approach against state-of-the-art methods across two categories: (1) Supervised Learning (SL) methods, including RetinexNet [59], MBLLEN [31], KindD [63], CUE [65], WaveletDiffusion [20], PyDiff [66], and Retinexformer [4]; and (2) Unsupervised/Zero-shot Learning (UL) methods, including Zero-DCE [15], Zero-DCE++ [25], EnlightenGAN [21], RUAS [29], SCI [33], pairLIE [12], CLIP-LIT [27], GDP [10], QuadPrior [54], and AGLLDiff [28].

Evaluation Metrics. For paired datasets, we employ three full-reference metrics PSNR and SSIM [57] for distortion assessment, and LPIPS [62] for perceptual quality evaluation. For unpaired datasets, we utilize three non-reference metrics TOPIQ [5], PI [2], DBCNN [34], and TReS [14] to measure enhancement quality without ground truth references.

Implementation Details. Our framework is implemented using PyTorch and evaluated on a single NVIDIA RTX 3090 GPU. For the generative prior, we utilize a pre-

Table 1. Quantitative comparisons on the paired datasets. The best results are highlighted in **bold**. ‘SL’ and ‘UL’ indicate that the methods belong to supervised and unsupervised methods, respectively.

Type	Method	LOLv1			LOLv2-synthetic			LOLv2-real captured			LSRW-Huawei			LSRW-Nikon		
		PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
SL	RetinexNet [58]	16.434	0.431	0.581	17.305	0.788	0.247	18.175	0.517	0.473	17.370	0.409	0.526	13.713	0.300	0.445
	MBLLEN [31]	18.901	0.787	0.367	16.451	0.723	0.290	19.597	0.777	0.404	18.528	0.562	0.519	14.732	0.431	0.408
	KinD [63]	15.042	0.600	0.508	17.786	0.781	0.281	20.785	0.848	0.283	17.272	0.487	0.493	15.653	0.452	0.336
	CUE [65]	23.262	0.821	0.339	17.156	0.765	0.334	23.156	0.821	0.321	19.234	0.571	0.473	16.423	0.470	0.367
	WaveletDiffusion [20]	25.725	0.867	0.298	24.733	0.877	0.203	21.133	0.838	0.285	18.689	0.414	0.510	16.123	0.346	0.445
	PyDiff [66]	26.453	0.865	0.172	24.433	0.895	0.172	20.803	0.833	0.367	19.433	0.525	0.497	15.702	0.425	0.370
	Retinexformer [4]	25.094	0.837	0.255	22.337	0.952	0.101	21.178	0.856	0.286	19.570	0.567	0.481	15.901	0.448	0.350
UL	Zero-DCE [15]	15.433	0.666	0.385	18.331	0.838	0.177	18.985	0.680	0.361	16.406	0.485	0.451	14.979	0.419	0.377
	Zero-DCE++ [25]	17.864	0.480	0.409	18.568	0.835	0.199	17.864	0.480	0.409	16.560	0.490	0.459	15.554	0.418	0.380
	EnlightenGAN [21]	18.287	0.725	0.516	16.987	0.805	0.391	20.045	0.737	0.482	18.933	0.522	0.536	15.784	0.417	0.421
	RUAS [29]	17.129	0.701	0.382	13.584	0.672	0.436	16.172	0.672	0.415	16.040	0.539	0.551	12.242	0.451	0.487
	SCI [33]	15.125	0.618	0.378	15.769	0.763	0.236	18.052	0.632	0.348	15.681	0.445	0.440	14.677	0.432	0.347
	PairLIE [12]	19.192	0.807	0.349	19.806	0.842	0.274	20.689	0.804	0.343	19.250	0.562	0.449	15.606	0.432	0.379
	CLIP-LIT [27]	12.810	0.587	0.587	16.333	0.794	0.200	15.708	0.635	0.371	13.453	0.433	0.460	13.401	0.384	0.374
	GDP [10]	16.404	0.603	0.412	11.924	0.480	0.295	14.851	0.551	0.436	14.002	0.385	0.494	11.386	0.312	0.401
	QuadPrior [54]	18.820	0.829	0.349	16.517	0.797	0.302	20.827	0.840	0.337	18.945	0.592	0.425	15.144	0.523	0.305
	AGLLDiff [28]	21.046	0.865	0.309	19.558	0.830	0.336	20.394	0.854	0.326	19.466	0.571	0.483	14.789	0.465	0.380
	Ours	21.175	0.876	0.293	20.222	0.849	0.299	21.260	0.858	0.276	19.693	0.595	0.472	15.914	0.478	0.365

Table 2. Quantitative comparisons on the unpaired datasets. Idem as Table 1.

Type	Method	DICM				LIME				MEF				NPE				VV			
		TOPIQ $\uparrow$	PI $\downarrow$	DBCNN $\uparrow$	TReS $\uparrow$	TOPIQ $\uparrow$	PI $\downarrow$	DBCNN $\uparrow$	TReS $\uparrow$	TOPIQ $\uparrow$	PI $\downarrow$	DBCNN $\uparrow$	TReS $\uparrow$	TOPIQ $\uparrow$	PI $\downarrow$	DBCNN $\uparrow$	TReS $\uparrow$	TOPIQ $\uparrow$	PI $\downarrow$	DBCNN $\uparrow$	TReS $\uparrow$
SL	RetinexNet [58]	0.506	3.237	0.537	68.697	0.528	4.233	0.595	74.811	0.580	3.779	0.600	80.740	0.547	3.207	0.611	82.294	0.483	4.867	0.562	72.021
	MBLLEN [31]	0.492	3.965	0.469	69.204	0.535	3.865	0.544	76.466	0.519	4.313	0.500	75.447	0.557	3.515	0.593	82.338	0.490	4.984	0.494	62.402
	KinD [63]	0.507	3.767	0.521	73.489	0.541	4.020	0.583	79.440	0.581	3.805	0.589	85.351	0.559	3.467	0.592	83.400	0.527	4.335	0.567	70.895
	CUE [65]	0.418	3.323	0.434	60.184	0.461	3.133	0.501	63.481	0.470	3.156	0.475	69.449	0.482	2.677	0.537	74.074	0.401	3.898	0.456	54.159
	WaveletDiffusion [20]	0.476	3.661	0.487	67.259	0.523	3.667	0.540	70.312	0.534	2.799	0.591	79.109	0.556	3.367	0.581	76.457	0.558	3.343	0.573	65.704
	PyDiff [66]	0.412	3.546	0.451	58.323	0.510	3.974	0.556	72.092	0.546	3.582	0.585	77.372	0.557	2.796	0.568	76.233	0.573	3.997	0.602	70.003
	Retinexformer [4]	0.430	3.078	0.433	59.458	0.516	2.973	0.562	72.543	0.550	3.102	0.539	76.160	0.545	2.547	0.544	75.626	0.591	3.367	0.616	69.331
UL	Zero-DCE [15]	0.465	3.152	0.474	65.704	0.511	3.197	0.551	72.444	0.576	3.147	0.584	78.041	0.556	2.877	0.600	80.239	0.434	3.334	0.510	60.616
	Zero-DCE++ [25]	0.451	3.152	0.463	63.431	0.504	3.091	0.541	71.599	0.565	3.244	0.575	76.564	0.549	3.101	0.585	77.850	0.422	3.279	0.484	56.943
	EnlightenGAN [21]	0.430	3.391	0.421	58.992	0.521	3.133	0.535	71.141	0.551	3.205	0.556	76.388	0.528	3.100	0.556	72.212	0.414	3.667	0.462	55.837
	RUAS [29]	0.373	5.173	0.418	52.780	0.446	4.178	0.480	61.329	0.453	5.012	0.487	66.984	0.439	5.479	0.487	68.572	0.315	4.313	0.460	41.090
	SCI [33]	0.431	3.702	0.484	58.720	0.490	3.233	0.535	67.173	0.554	3.279	0.576	73.479	0.505	3.505	0.560	76.673	0.392	3.607	0.471	51.130
	PairLIE [12]	0.437	3.667	0.437	63.976	0.483	3.551	0.516	71.330	0.522	3.023	0.538	78.751	0.502	3.033	0.543	75.326	0.421	3.677	0.480	60.887
	CLIP-LIT [27]	0.507	3.233	0.518	68.444	0.528	3.122	0.569	72.311	0.602	3.197	0.598	79.005	0.567	2.878	0.613	79.278	0.482	3.071	0.562	63.997
	GDP [10]	0.454	3.577	0.447	62.190	0.512	3.655	0.535	70.486	0.539	3.467	0.551	74.388	0.545	3.011	0.564	77.012	0.463	3.167	0.497	57.870
	QuadPrior [54]	0.454	3.156	0.478	64.726	0.442	3.257	0.452	61.730	0.390	3.347	0.404	55.401	0.461	2.967	0.453	64.522	0.524	3.241	0.555	73.447
	AGLLDiff [28]	0.499	3.072	0.503	61.002	0.546	2.977	0.563	79.450	0.578	3.179	0.573	85.970	0.563	2.677	0.578	81.743	0.501	2.782	0.526	68.355
	Ours	0.508	3.034	0.542	76.184	0.549	3.074	0.597	79.766	0.583	2.710	0.601	86.659	0.568	2.801	0.614	84.481	0.568	2.933	0.567	73.501

trained 256×256 diffusion denoising model trained on the ImageNet dataset [7], as provided by [8]. The number of denoising steps is set to the default value of 200. It costs around 1.72 minute to render a 400×600 resolution image using 2.3GB memory. The weighting coefficients for the loss functions are empirically set to $\lambda_1 = 1$, $\lambda_2 = 0.5$, $\lambda_3 = 0.5$, and $\lambda_4 = 0.01$.

4.2. Comparison with Existing Methods

Quantitative Comparison. Tables 1 and 2 present a comprehensive quantitative comparison against recent state-of-the-art methods on both paired and unpaired low-light benchmarks. Across all datasets, our method demonstrates consistently superior performance, achieving a balanced improvement in both full-reference and no-reference metrics while operating in a training-free, unsupervised manner. In particular, on paired datasets, our method achieves consistently higher scores in full-reference metrics (e.g., PSNR and SSIM), indicating superior structural fidelity and significantly reduced artifacts compared to recent diffusion-based baselines (e.g., AGLLDiff [28] and GDP [10]). Simultaneously, our framework demonstrates exceptional per-

formance on unpaired real-world datasets (Table 2) evaluated by non-reference metrics, suggesting that our restored images better align with natural human visual perception. Furthermore, our method remarkably outperforms recent supervised approaches (e.g., PyDiff [66] and Retinexformer [4]) on real-world datasets like LOLv2-real [60] and LSRW-Huawei [17], demonstrating more robust generalization capabilities.

Qualitative Comparison. Visual comparisons on challenging low-light scenes are presented in Fig. 4. Methods such as MBLLEN [31], Zero-DCE++ [25], RUAS [29], and SCI [33] struggle to adequately recover illumination, often leaving the outputs severely under-exposed. Other baselines, like EnlightenGAN [21] and CUE [65], manage to brighten the images but introduce severe color distortion and noticeably amplify background noise. When examining recent diffusion-based baselines, GDP [10] exhibits severe over-smoothing, heavily degrading fine textures and losing crucial high-frequency details, as clearly visible in the brick wall and the sculpture’s facial features. While QuadPrior [54] and AGLLDiff [28] improve overall brightness, they frequently compromise structural fidelity, pro-



Figure 5. Visual comparison of high-frequency structures using Sobel edge maps.

ducing distorted shapes, hallucinating unnatural edges, and introducing color shifts. Fig. 5 further visualizes the recovered high-frequency structures using Sobel edge maps, our proposed method reconstructs sharper boundaries, such as the straight lines of the background brick wall and the fine-grained facial contours of the sculpture, avoiding the blurring and hallucination artifacts in previous direct-generation methods. In summary, our proposed method restores both structural integrity and accurate illumination more faithfully to the ground truth. As highlighted in the zoomed-in patches in Fig. 4, by leveraging the decoupled noiseless structural proxy and the ISP-inspired physical recovery, our approach simultaneously produces much sharper textures, natural color, and proper exposure. Ultimately, it achieves a satisfactory overall appearance, avoiding the blurring and hallucination artifacts that plague existing direct-generation diffusion models.

P_t	α	γ	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
$\times$	$\times$	$\times$	10.872	0.428	0.391
$\checkmark$	$\times$	$\times$	11.527	0.485	0.386
$\checkmark$	$\checkmark$	$\times$	15.819	0.739	0.348
$\checkmark$	$\times$	$\checkmark$	19.989	0.834	0.317
$\checkmark$	$\checkmark$	$\checkmark$	20.222	0.849	0.299

Table 3. Ablation studies on LOLv2-synthetic dataset.

4.3. Ablation Study

To validate the effectiveness of our proposed core components, we conduct comprehensive ablation studies on the LOLv2-synthetic dataset. As detailed in Table 3, we progressively isolate the contributions of the noiseless structural proxy (P_t), the adaptive light gain (α), and the non-linear gamma corrector (γ).

Effectiveness of the Structural Proxy. The baseline model (Row 1), which directly generates the enhanced image without our physical decoupling strategy, yields extremely poor quantitative results due to severe structural degradation and high prediction uncertainty. By introducing solely the noiseless structural proxy (Row 2), we observe slight improvements in three metrics. Note that because the proxy intentionally isolates structural denoising from illumination recovery, the resulting images remain physically under-exposed. Consequently, while the structural details is

generated, the absolute PSNR remains relatively low, confirming the absolute necessity of subsequent operations.

Effectiveness of ISP-Inspired Recovery. Building upon the structural proxy, we evaluate the individual impacts of our ISP-inspired recovery operations. When introducing only the linear light gain α (Row 3), the performance surges significantly (PSNR improves from 11.527 dB to 15.819 dB, SSIM to 0.739), demonstrating the important role of compensating for the low-photon deficit. Moreover, employing only the non-linear gamma corrector γ (Row 4) yields an even more dramatic improvement (PSNR reaches 19.989 dB), highlighting its superior capability in optimal exposure remapping and global contrast balancing.

The full model (Row 5) integrates both light gain α and exposure corrector γ on top of the proxy foundation, achieves the best overall performance, thoroughly validating our physically-guided joint optimization strategy.

5. Conclusion and Limitation

In this work, we present π -Diff, an unsupervised, training-free framework based on diffusion prior for LLIE. We address the critical challenges of existing direct generation paradigm by introducing a noiseless structural proxy to maintain fine-grained details, and a physical ISP-inspired mechanism to recover illumination through learnable light gain and gamma corrector. Our framework successfully integrates real-world camera imaging priors into the generative reverse process without any paired training data. Our experiments have demonstrated that π -Diff achieves exceptional performance on public datasets in detail preservation, color restoration, and illumination recovery.

Limitations and Future Work. While our proposed approach achieves superior enhancement performance, it needs a relatively long inference time per image. This limitation arises because accurately estimating the dynamic enhancement parameters (i.e., light gain α and gamma corrector γ) relies on a sufficient number of diffusion sampling steps. Employing extremely few steps can lead to inaccurate estimation and compromise the final quality. In future work, we plan to accelerate the overall enhancement process by integrating advanced diffusion acceleration techniques (e.g., efficient ODE solvers or distillation) and designing more lightweight architectures for the parameter prediction modules, thereby reducing inference latency while preserving the high-fidelity, physically-guided restoration capabilities.

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References

- [1] Titas Anciukevičius, Zexiang Xu, Matthew Fisher, Paul Henderson, Hakan Bilen, Niloy J Mitra, and Paul Guerrero. Renderdiffusion: Image diffusion for 3d reconstruction, inpainting and generation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 12608–12618, 2023. 3
- [2] Yochai Blau, Roey Mechrez, Radu Timofte, Tomer Michaeli, and Lihi Zelnik-Manor. The 2018 pirm challenge on perceptual image super-resolution. In *Proceedings of the European conference on computer vision (ECCV) workshops*, pages 0–0, 2018. 6
- [3] Alan C Bovik. *Handbook of image and video processing*. Academic press, 2010. 2
- [4] Yuanhao Cai, Hao Bian, Jing Lin, Haoqian Wang, Radu Timofte, and Yulun Zhang. Retinexformer: One-stage retinex-based transformer for low-light image enhancement. *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 12470–12479, 2023. 6, 7
- [5] Chaofeng Chen, Jiadi Mo, Jingwen Hou, Haoning Wu, Liang Liao, Wenxiu Sun, Qiong Yan, and Weisi Lin. Topiq: A top-down approach from semantics to distortions for image quality assessment. *IEEE Transactions on Image Processing*, 33:2404–2418, 2024. 6
- [6] Hyungjin Chung, Jeongsol Kim, Michael T. McCann, Marc Louis Klasky, and J. C. Ye. Diffusion posterior sampling for general noisy inverse problems. *ArXiv*, abs/2209.14687, 2022. 4
- [7] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, K. Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. *2009 IEEE Conference on Computer Vision and Pattern Recognition*, pages 248–255, 2009. 7
- [8] Prafulla Dhariwal and Alexander Nichol. Diffusion models beat gans on image synthesis. *Advances in neural information processing systems*, 34:8780–8794, 2021. 2, 4, 7
- [9] Bradley Efron. Tweedie’s formula and selection bias. *Journal of the American Statistical Association*, 106:1602 – 1614, 2011. 4
- [10] Ben Fei, Zhaoyang Lyu, Liang Pan, Junzhe Zhang, Weidong Yang, Tianyue Luo, Bo Zhang, and Bo Dai. Generative diffusion prior for unified image restoration and enhancement. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 9935–9946, 2023. 2, 3, 6, 7
- [11] Xueyang Fu, Delu Zeng, Yue Huang, Xiao-Ping Zhang, and Xinghao Ding. A weighted variational model for simultaneous reflectance and illumination estimation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2782–2790, 2016. 2
- [12] Zhenqi Fu, Yan Yang, Xiaotong Tu, Yue Huang, Xinghao Ding, and Kai-Kuang Ma. Learning a simple low-light image enhancer from paired low-light instances. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 22252–22261, 2023. 1, 2, 6, 7
- [13] Sicheng Gao, Xuhui Liu, Bohan Zeng, Sheng Xu, Yanjing Li, Xiaoyan Luo, Jianzhuang Liu, Xiantong Zhen, and Baochang Zhang. Implicit diffusion models for continuous super-resolution. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10021–10030, 2023. 3
- [14] S Alireza Golestaneh, Saba Dadsetan, and Kris M Kitani. No-reference image quality assessment via transformers, relative ranking, and self-consistency. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, pages 1220–1230, 2022. 6
- [15] Chunle Guo, Chongyi Li, Jichang Guo, Chen Change Loy, Junhui Hou, Sam Kwong, and Runmin Cong. Zero-reference deep curve estimation for low-light image enhancement. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 1780–1789, 2020. 2, 3, 6, 7
- [16] Xiaojie Guo, Yu Li, and Haibin Ling. Lime: Low-light image enhancement via illumination map estimation. *IEEE Transactions on image processing*, 26(2):982–993, 2016. 2, 6
- [17] Jiang Hai, Zhu Xuan, Ren Yang, Yutong Hao, Fengzhu Zou, Fang Lin, and Songchen Han. R2rnet: Low-light image enhancement via real-low to real-normal network. *Journal of Visual Communication and Image Representation*, 90: 103712, 2023. 6, 7
- [18] Khurram Azeem Hashmi, Goutham Kallempudi, Didier Stricker, and Muhammad Zeshan Afzal. Featenhancer: Enhancing hierarchical features for object detection and beyond under low-light vision. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 6725–6735, 2023. 2
- [19] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33:6840–6851, 2020. 2
- [20] Hailin Jiang, Ao Luo, Songchen Han, Haoqiang Fan, and Shuaicheng Liu. Low-light image enhancement with wavelet-based diffusion models. *ACM Transactions on Graphics (TOG)*, 42:1 – 14, 2023. 6, 7
- [21] Yifan Jiang, Xinyu Gong, Ding Liu, Yu Cheng, Chen Fang, Xiaohui Shen, Jianchao Yang, Pan Zhou, and Zhangyang Wang. Enlightengan: Deep light enhancement without paired supervision. *IEEE transactions on image processing*, 30:2340–2349, 2021. 1, 2, 6, 7
- [22] Yunwei Lan, Zhigao Cui, Chang Liu, Jialun Peng, Nian Wang, Xin Luo, and Dong Liu. Exploiting diffusion prior for real-world image dehazing with unpaired training. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 4455–4463, 2025. 3
- [23] Edwin H Land. The retinex theory of color vision. *Scientific american*, 237(6):108–129, 1977. 2
- [24] Chulwoo Lee, Chul Lee, and Chang-Su Kim. Contrast enhancement based on layered difference representation of 2d histograms. *IEEE transactions on image processing*, 22(12): 5372–5384, 2013. 6

- [25] Chongyi Li, Chunle Guo, and Chen Change Loy. Learning to enhance low-light image via zero-reference deep curve estimation. *IEEE transactions on pattern analysis and machine intelligence*, 44(8):4225–4238, 2021. 1, 2, 6, 7
- [26] Jinlong Li, Baolu Li, Zhengzhong Tu, Xinyu Liu, Qing Guo, Felix Juefei-Xu, Runsheng Xu, and Hongkai Yu. Light the night: A multi-condition diffusion framework for unpaired low-light enhancement in autonomous driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 15205–15215, 2024. 2
- [27] Zhexin Liang, Chongyi Li, Shangchen Zhou, Ruicheng Feng, and Chen Change Loy. Iterative prompt learning for unsupervised backlit image enhancement. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 8094–8103, 2023. 1, 6, 7
- [28] Yunlong Lin, Tian Ye, Sixiang Chen, Zhenqi Fu, Yingying Wang, Wenhao Chai, Zhaoxu Xing, Wenxue Li, Lei Zhu, and Xinghao Ding. Agldiff: Guiding diffusion models towards unsupervised training-free real-world low-light image enhancement. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 5307–5315, 2025. 2, 3, 6, 7
- [29] Risheng Liu, Long Ma, Jiaao Zhang, Xin Fan, and Zhongxuan Luo. Retinex-inspired unrolling with cooperative prior architecture search for low-light image enhancement. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10561–10570, 2021. 1, 2, 6, 7
- [30] Andreas Lugmayr, Martin Danelljan, Andres Romero, Fisher Yu, Radu Timofte, and Luc Van Gool. Repaint: Inpainting using denoising diffusion probabilistic models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 11461–11471, 2022. 3
- [31] Feifan Lv, Feng Lu, Jianhua Wu, and Chongsoon Lim. Mbllen: Low-light image/video enhancement using cnns. In *Bmvc*, page 4. Northumbria University, 2018. 2, 6, 7
- [32] Kede Ma, Kai Zeng, and Zhou Wang. Perceptual quality assessment for multi-exposure image fusion. *IEEE Transactions on Image Processing*, 24(11):3345–3356, 2015. 6
- [33] Long Ma, Tengyu Ma, Risheng Liu, Xin Fan, and Zhongxuan Luo. Toward fast, flexible, and robust low-light image enhancement. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 5637–5646, 2022. 1, 2, 6, 7
- [34] A Deep Bilinear Convolutional Neural Network. Blind image quality assessment using a deep bilinear convolutional neural network. *Deep Bilinear Convolutional Neural*, 2022. 6
- [35] Chen Hee Ooi and Nor Ashidi Mat Isa. Quadrants dynamic histogram equalization for contrast enhancement. *IEEE Transactions on Consumer Electronics*, 56(4):2552–2559, 2010. 2
- [36] Long Peng, Wenbo Li, Renjing Pei, Jingjing Ren, Jiaqi Xu, Yang Wang, Yang Cao, and Zheng-Jun Zha. Towards realistic data generation for real-world super-resolution. In *The Thirteenth International Conference on Learning Representations*. 3
- [37] Long Peng, Yang Cao, Renjing Pei, Wenbo Li, Jiaming Guo, Xueyang Fu, Yang Wang, and Zheng-Jun Zha. Efficient real-world image super-resolution via adaptive directional gradient convolution. *arXiv preprint arXiv:2405.07023*, 2024.
- [38] Long Peng, Yang Cao, Yuejin Sun, and Yang Wang. Lightweight adaptive feature de-drifting for compressed image classification. *IEEE Transactions on Multimedia*, 26: 6424–6436, 2024.
- [39] Long Peng, Wenbo Li, Jiaming Guo, Xin Di, Haoze Sun, Yong Li, Renjing Pei, Yang Wang, Yang Cao, and Zheng-Jun Zha. Unveiling hidden details: A raw data-enhanced paradigm for real-world super-resolution. *arXiv preprint arXiv:2411.10798*, 2024.
- [40] Long Peng, Xin Di, Zhanfeng Feng, Wenbo Li, Renjing Pei, Yang Wang, Xueyang Fu, Yang Cao, and Zheng-Jun Zha. Directing mamba to complex textures: An efficient texture-aware state space model for image restoration. *arXiv preprint arXiv:2501.16583*, 2025.
- [41] Long Peng, Yang Wang, Xin Di, Xueyang Fu, Yang Cao, Zheng-Jun Zha, et al. Boosting image de-raining via central-surrounding synergistic convolution. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 6470–6478, 2025.
- [42] Long Peng, Anran Wu, Wenbo Li, Peizhe Xia, Xueyuan Dai, Xinjie Zhang, Xin Di, Haoze Sun, Renjing Pei, Yang Wang, et al. Pixel to gaussian: Ultra-fast continuous super-resolution with 2d gaussian modeling. *arXiv preprint arXiv:2503.06617*, 2025. 3
- [43] Robin Rombach, A. Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10674–10685, 2021. 2
- [44] Chitwan Saharia, William Chan, Huiwen Chang, Chris A. Lee, Jonathan Ho, Tim Salimans, David J. Fleet, and Mohammad Norouzi. Palette: Image-to-image diffusion models. *ACM SIGGRAPH 2022 Conference Proceedings*, 2021. 4
- [45] Tim Salimans and Jonathan Ho. Progressive distillation for fast sampling of diffusion models. In *International Conference on Learning Representations*. 4
- [46] Shuyao Shang, Zhengyang Shan, Guangxing Liu, LunQian Wang, XingHua Wang, Zekai Zhang, and Jinglin Zhang. Resdiff: Combining cnn and diffusion model for image super-resolution. In *Proceedings of the AAAI conference on artificial intelligence*, pages 8975–8983, 2024. 3
- [47] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. *arXiv preprint arXiv:2010.02502*, 2020. 2
- [48] Yi Tang, Hiroshi Kawasaki, and Takafumi Iwaguchi. Underwater image enhancement by transformer-based diffusion model with non-uniform sampling for skip strategy. In *Proceedings of the 31st ACM international conference on multimedia*, pages 5419–5427, 2023. 3
- [49] Pascal Vincent. A connection between score matching and denoising autoencoders. *Neural Computation*, 23:1661–1674, 2011. 4
- [50] Vassilios Vonikakis, Rigas Kouskouridas, and Antonios Gasteratos. On the evaluation of illumination compensation

- algorithms. *Multimedia Tools and Applications*, 77(8):9211–9231, 2018. 6
- [51] Min Wang, Xin Huang, and Qing Wang. Atm-nerf: Learning adaptive tone mapping for normal-light neural radiance field reconstruction. *2025 IEEE International Conference on Multimedia and Expo (ICME)*, pages 1–6, 2025. 2
- [52] Min Wang, Xin Huang, Guoqing Zhou, Qifeng Guo, and Qing Wang. Bright-nerf: Brightening neural radiance field with color restoration from low-light raw images. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 7817–7825, 2025. 2
- [53] Shuhang Wang, Jin Zheng, Hai-Miao Hu, and Bo Li. Naturalness preserved enhancement algorithm for non-uniform illumination images. *IEEE transactions on image processing*, 22(9):3538–3548, 2013. 6
- [54] Wenjing Wang, Huan Yang, Jianlong Fu, and Jiaying Liu. Zero-reference low-light enhancement via physical quadruple priors. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 26057–26066, 2024. 3, 6, 7
- [55] Yang Wang, Long Peng, Liang Li, Yang Cao, and Zheng-Jun Zha. Decoupling-and-aggregating for image exposure correction. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 18115–18124, 2023. 1
- [56] Yufei Wang, Wenhan Yang, Xinyuan Chen, Yaohui Wang, Lanqing Guo, Lap-Pui Chau, Ziwei Liu, Yu Qiao, Alex C Kot, and Bihan Wen. Sinsr: diffusion-based image super-resolution in a single step. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 25796–25805, 2024. 3
- [57] Zhou Wang, Alan C Bovik, Hamid R Sheikh, and Eero P Simoncelli. Image quality assessment: from error visibility to structural similarity. *IEEE transactions on image processing*, 13(4):600–612, 2004. 6
- [58] Chen Wei, Wenjing Wang, Wenhan Yang, and Jiaying Liu. Deep retinex decomposition for low-light enhancement. 5, 7
- [59] Chen Wei, Wenjing Wang, Wenhan Yang, and Jiaying Liu. Deep retinex decomposition for low-light enhancement. *arXiv preprint arXiv:1808.04560*, 2018. 2, 6
- [60] Wenhan Yang, Wenjing Wang, Haofeng Huang, Shiqi Wang, and Jiaying Liu. Sparse gradient regularized deep retinex network for robust low-light image enhancement. *IEEE Transactions on Image Processing*, 30:2072–2086, 2021. 6, 7
- [61] Shibai Yin, Yiwei Shi, Yibin Wang, and Yee-Hong Yang. When aware haze density meets diffusion model for synthetic-to-real dehazing. *IEEE Transactions on Circuits and Systems for Video Technology*, 35(5):4242–4255, 2025. 3
- [62] Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 586–595, 2018. 6
- [63] Yonghua Zhang, Jiawan Zhang, and Xiaojie Guo. Kindling the darkness: A practical low-light image enhancer. In *Proceedings of the 27th ACM international conference on multimedia*, pages 1632–1640, 2019. 2, 6, 7
- [64] Chen Zhao, Chenyu Dong, Weiling Cai, and Yueyue Wang. Learning a physical-aware diffusion model based on transformer for underwater image enhancement. *IEEE Transactions on Geoscience and Remote Sensing*, 2026. 3
- [65] Naishan Zheng, Man Zhou, Yanmeng Dong, Xiangyu Rui, Jie Huang, Chongyi Li, and Feng Zhao. Empowering low-light image enhancer through customized learnable priors. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 12559–12569, 2023. 2, 6, 7
- [66] Dewei Zhou, Zongxin Yang, and Yi Yang. Pyramid diffusion models for low-light image enhancement. *ArXiv*, abs/2305.10028, 2023. 6, 7